PCF-2: A Program Statement for the Cubic Extension of the Δ-Discriminant Transcendence Predicate

Version 1.1

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Abstract

This paper is a *program statement*, not a results paper. It frames PCF-2 (P13 in the SIARC stack), the cubic extension of PCF-1 [1] from degree-two polynomial continued fractions to degree-three. PCF-1 established a sharp empirical dichotomy across 30 degree-two families: the sign of the balanced discriminant $\Delta_2 = \beta^2 - 4\alpha\gamma$ predicts whether the limit of the PCF admits an elementary closed form ($\Delta_2 > 0$, 24/24 Trans-stratum families) or fails PSLQ detection against an 18-element basis at 220 digits ($\Delta_2 < 0$, 6/6 CM candidates), with the Stokes-exponent predicate $S < 1$ confirmed across all six $\Delta < 0$ families. The cubic case is qualitatively different. The natural invariant of the denominator polynomial $b(n) = \alpha_3 n^3 + \alpha_2 n^2 + \alpha_1 n + \alpha_0$ is the cubic discriminant Δ_3, which is no longer a sign invariant but a quantity whose sign *and* Galois-group structure (S_3, $A_3 = C_3$, or split) jointly determine the splitting field and the analytic obstruction. We propose four open conjectures: (i) a cubic analogue of the sign dichotomy with the Galois-group axis; (ii) a Galois-stratified CM-field structure replacing the binary imaginary-quadratic predicate; (iii) a Stokes signal calibrated for irreducible-cubic-CM families; and (iv) an alignment with the Painlevé IV equation rather than P-III, motivated by the cubic Galois-group / Painlevé hierarchy correspondence. We sketch a four-to-six-week computational plan that reuses the infrastructure of PCF-1 and identifies three concrete first sessions (PCF-2 A/B/C). PCF-2 is positioned to either confirm that the Δ-predicate of PCF-1 extends to higher degree (universality) or to identify degree two as a special boundary (specificity). We further state Conjecture B4, an unsplit sharp form of the WKB exponent identity for degree-d PCFs in scope, asserting $A = 2d$, and verify it on 50/50 cubic families (mean $A_{\text{fit}} = 5.978$, $\sigma = 0.026$). A bin-driven refinement Conjecture B4′ was tested and empirically falsified at $d = 3$, ruling out the trichotomy bin as the controlling invariant for A and motivating a "$d = 2$ anomaly" open problem on why degree 2 admits an analogous sgn(Δ)-driven split $A \in \{3, 4\}$ (Theorem 5 of PCF-1 v1.3) while degree 3 does not.

Series label: P13 / PCF-2. This program statement is P13 in the AI Epistemic Governance Stack and simultaneously PCF-2 in the mathematical companion track. It anchors a research program; it does not contain new computational results. The companion document is the SIARC umbrella program statement [2].

*ORCID: 0009-0000-6192-8273. Independent Researcher, Yokohama, Japan. **AI use disclosure.** This program statement was prepared using the AEAL/SIARC multi-agent methodology. GitHub Copilot (powered by Anthropic Claude Opus 4.7) and Anthropic Claude were used to assist with literature synthesis, drafting of expository prose, and consistency-checking against prior preprints in the series. The v1.1 update integrates the empirical harvest of Sessions A2 (conductor-7 anchor), B (PSLQ scan of the $-_S_3_$CM bin and the WKB-exponent fit), and C1 (non-CM cubic WKB completion). All conjecture formulations, scope decisions, and editorial choices were made by the human author, who takes full responsibility for the content. The AI tools are not authors. See the AEAL methodology of [6].

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1 Position in the SIARC Stack

The SIARC research program organizes polynomial continued fractions (PCFs) into an arithmetic stratification indexed by the dominant growth degree and an obstruction tier. The program-level architecture is articulated in the umbrella program statement [2]; the $d = 2$ instantiation is PCF-1 [1]; the present document opens the $d = 3$ instantiation.

What PCF-1 established. For PCFs of the form

$$K(a_n, b_n) = b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cdots}}$$

with $a_n \in \mathbb{Z}[n]_{\leq 1}$ and $b_n \in \mathbb{Z}[n]_{\leq 2}$, PCF-1 proposed the sign of the balanced discriminant $\Delta_2 = \beta^2 - 4\alpha\gamma$ as a transcendence predicate. The core empirical finding was a sharp dichotomy across 30 families: 24/24 Trans-stratum families with $\Delta_2 \in \{+1, +8\}$ admit elementary closed forms via Lindemann–Weierstrass, while all 6/6 CM candidates with $\Delta_2 \in \{-3, -4, -7, -11, -20, -28\}$ return PSLQ non-detection against an 18-element basis at working precision 220 digits, and all six exhibit the Stokes-exponent signature $S < 1$ under a CM-respecting deformation. The V_quad family ($\Delta_2 = -11$) is the explicit Painlevé III(D_6) prototype. PCF-1 v1.2 also showed (via QL15, $\Delta_2 = -20$, class number $h = 2$, non-Heegner; and the intra-field pair QL06/QL26 in $\mathbb{Q}(\sqrt{-7})$ at $h = 1$ versus $h = 2$) that the controlling invariant is the imaginary-quadratic field $\mathbb{Q}(\sqrt{\Delta})$ itself, not its class number [1].

Why degree three is the natural next step. Three structural features make $d = 3$ the canonical extension target.

1. *Discriminant lifts naturally.* The cubic discriminant Δ_3 of $b(n) = \alpha_3 n^3 + \alpha_2 n^2 + \alpha_1 n + \alpha_0$ is a classical $\mathrm{SL}_2(\mathbb{Z})$-invariant of degree 4 in the coefficients (Definition 1). Its sign distinguishes totally real cubic splitting from one-real-two-complex splitting, the direct analogue of the $d = 2$ sign dichotomy.

2. *Galois group is no longer trivial.* A degree-three irreducible polynomial has Galois group S_3 or $A_3 = C_3$ over $\mathbb{Q}$, distinguished by whether Δ_3 is a square in $\mathbb{Q}$. This adds an arithmetic axis absent in the degree-two case (where $\mathrm{Gal}(b/\mathbb{Q})$ is C_2 whenever b is irreducible).

3. *Painlevé hierarchy aligns with cubic Galois structure.* The classical Painlevé equations admit a cubic-singularity analogue (P-IV) whose monodromy data is naturally indexed by cubic Galois groups, in the same way P-III governs the imaginary-quadratic CM case in PCF-1.

What PCF-2 will not address. PCF-2 does not extend automatically to degree four or higher. The cubic case is the unique additional case in which the Galois group is finite and explicitly classifiable, and in which the Painlevé correspondence is well-posed. Extensions to $d = 4$ and beyond are deferred to PCF-3 and later instantiations of the SIARC program.

2 Setup: Degree-Three Polynomial CFs

We work with PCFs of the form

$$K(a_n, b_n) = b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cdots}} \tag{1}$$

with

$$a_n = \delta_1\, n + \delta_0, \qquad \delta_1, \delta_0 \in \mathbb{Z}, \tag{2}$$

$$b_n = \alpha_3\, n^3 + \alpha_2\, n^2 + \alpha_1\, n + \alpha_0, \qquad \alpha_i \in \mathbb{Z},\ \alpha_3 \neq 0. \tag{3}$$

The convergents satisfy the Wallis recurrence $p_n = b_n p_{n-1} + a_n p_{n-2}$ with the seeds $(p_{-1}, p_0, q_{-1}, q_0) = (1, b_0, 0, 1)$, identical to the $d = 2$ convention of PCF-1 to facilitate code reuse. Coefficient ordering for the family b_n is $[\alpha_3, \alpha_2, \alpha_1, \alpha_0]$ (leading first), matching the PCF-1 `f1_base_computation.py` convention.

Definition 1 (Cubic balanced discriminant)**.** For a balanced PCF in the $d = 3$ class with b_n as in (3),

$$\Delta_3 \;=\; \mathrm{disc}(b) \;=\; 18\alpha_3\alpha_2\alpha_1\alpha_0 - 4\alpha_2^3\alpha_0 + \alpha_2^2\alpha_1^2 - 4\alpha_3\alpha_1^3 - 27\alpha_3^2\alpha_0^2.$$

We say a degree-three PCF is $\mathbb{Z}$-*primitive* if $\gcd(\alpha_3, \alpha_2, \alpha_1, \alpha_0) = 1$. We say it is *irreducible* if b is irreducible over $\mathbb{Q}$, and we always restrict to the irreducible $\mathbb{Z}$-primitive case in this program.

Remark 2. The PCF-1 normalization $\gcd(\alpha, \beta, \gamma) = 1$ extended to $d = 3$ removes the 1-parameter integer-scaling redundancy and forces Δ_3 to be a $\mathbb{Z}$-valued invariant of the family, not a class representative. The cubic discriminant is invariant under $\alpha_i \mapsto \alpha_i + c\binom{i}{1}$-style n-shifts only up to a predictable scaling; the convention (3) fixes the n-coordinate.

Splitting types and Galois groups. For an irreducible $\mathbb{Z}$-primitive cubic $b \in \mathbb{Z}[n]$:

- If $\Delta_3 > 0$, then b has three real roots and the splitting field is the cubic resolvent $\mathbb{Q}(\theta)$ for θ a real root of b, together with $\sqrt{\Delta_3} \in \mathbb{R}$. The Galois group is $\mathrm{Gal}(b/\mathbb{Q}) = A_3 = C_3$ when Δ_3 is a perfect square in $\mathbb{Q}$, and $\mathrm{Gal}(b/\mathbb{Q}) = S_3$ otherwise.

- If $\Delta_3 < 0$, then b has one real root and a complex conjugate pair, the splitting field contains $\mathbb{Q}(\sqrt{\Delta_3})$ (an imaginary-quadratic subfield), and the Galois group is necessarily $\mathrm{Gal}(b/\mathbb{Q}) = S_3$.

This trichotomy — A_3 totally real, S_3 totally real, S_3 complex — is the cubic analogue of the binary $\Delta_2 > 0$ vs. $\Delta_2 < 0$ split, and it is what gives PCF-2 genuinely new content beyond PCF-1.

Spec(K) in the cubic case. We retain the v5 schema of PCF-1,

$$\mathrm{Spec}(K) = (d, \Lambda, \Delta, \rho, \tau_{\mathrm{mono}}, \tau_{\mathrm{arith}}; S, \sigma),$$

with $d = 3$ now, and with $\Delta = \Delta_3$ as defined above. The Poincaré–Perron normalized growth rate Λ in the balanced $d = 3$ class satisfies the cubic

$$\Lambda^3 - \alpha_3\Lambda^2 - \alpha_3\delta_1\Lambda - \alpha_3\delta_0 = 0, \tag{4}$$

or a closely related cubic depending on the precise scaling convention; the analytic determination of (4) is part of PCF-2 Session A. The arithmetic sector τ_{arith} now records the splitting field of b together with the Galois group $\mathrm{Gal}(b/\mathbb{Q}) \in \{C_3, S_3\}$.

3 Predictions and Open Conjectures

We state four open conjectures, ordered from most direct lift of PCF-1 to most speculative.

Conjecture 3 (B_3 part (i): cubic sign dichotomy)**.** *Let K be a balanced $d = 3$ PCF with irreducible $\mathbb{Z}$-primitive denominator b as in (3), and let $\Delta_3 = \mathrm{disc}(b)$.*

(A) *If $\Delta_3 < 0$ (one real root, $\mathrm{Gal}(b/\mathbb{Q}) = S_3$), then $L = K(a_n, b_n)$ admits no elementary closed form: equivalently, $L \notin \mathbb{Q}(\pi, e, \log \mathbb{Q}^\times, \zeta(3), \sqrt{\mathbb{Q}}, \gamma)$ in the sense of PSLQ non-detection against the PCF-1 18-element basis at working precision ≥ 220 digits and integer bound $\geq 10^{10}$.*

(B) *If $\Delta_3 > 0$ and $\sqrt{\Delta_3} \in \mathbb{Q}$ (so $\mathrm{Gal}(b/\mathbb{Q}) = C_3$, the totally-real cyclic case), then L may admit a closed form; this is the cubic analogue of the PCF-1 $\Delta_2 > 0$ Trans-stratum and is the part of the conjecture most likely to admit explicit Lindemann–Weierstrass evaluation.*

(C) *If $\Delta_3 > 0$ and $\sqrt{\Delta_3} \notin \mathbb{Q}$ (so $\mathrm{Gal}(b/\mathbb{Q}) = S_3$, the totally-real non-cyclic case), then L admits no elementary closed form, with the same PSLQ-non-detection criterion as (A).*

Remark 4. Conjecture 3 is a *trichotomy* rather than a sign dichotomy. The new ingredient is the Galois-group axis: the cyclic case C_3 is the sole closed-form-admitting cell. This is the structural prediction PCF-2 Sessions A and B are designed to falsify or confirm.

Empirical evidence for B_3(i) (Session B). PCF-2 Session B (2026-05-01) ran a systematic PSLQ scan of the $-_S_3_$CM bin of the PCF-2 cubic-family catalogue (see §7 for the catalogue construction): 37 irreducible $\mathbb{Z}$-primitive cubics with $\Delta_3 < 0$ and $\mathrm{Gal}(b/\mathbb{Q}) = S_3$. Each L_b was probed at PSLQ tolerance 10^{-35} (working precision dps = 80) against a four-tier basis: (a) the PCF-1 18-element transcendental basis [1]; (b) the Γ-reflection block $\{\Gamma(k/n) : k = 1, \ldots, n-1\}$ for $n \in \{3, 4, 6\}$; (c) the cyclotomic ring of integers $\mathbb{Z}[\zeta_n]$ for $n \in \{3, 4, 5, 6, 7, 8, 9, 12\}$; and (d) Dirichlet $L(\chi, 1)$ values for the small-conductor real and quadratic characters. The result was 37/37 **BIN-CONSISTENT,** 0 **BIN-VIOLATING,** 0 **STALLED**: no nontrivial integer relation involving L_b was detected on any of the four bases for any family, and the cross-family pairwise sweep ($\binom{37}{2} = 666$ pairs) was likewise NULL. This is the first systematic positive evidence for Conjecture 3(A) at $d = 3$. Per-family records and AEAL claims are committed to `sessions/2026-05-01/PCF2-SESSION-B/` on the SIARC bridge.

Conjecture 5 (B_3 part (ii): Galois-stratified CM-field structure)**.** *Let K be a balanced $d = 3$ PCF with irreducible $\mathbb{Z}$-primitive denominator b. Let $\theta_1, \theta_2, \theta_3 \in \mathbb{C}$ be the roots of b, and let $L_b/\mathbb{Q}$ be the splitting field.*

1. *If $\Delta_3 < 0$, then $L_b = \mathbb{Q}(\theta, \sqrt{\Delta_3})$ where θ is the unique real root, $[L_b : \mathbb{Q}] = 6$, and the imaginary quadratic subfield $\mathbb{Q}(\sqrt{\Delta_3}) \subset L_b$ controls the Stokes data.*

2. *If $\Delta_3 > 0$ and $\mathrm{Gal}(b/\mathbb{Q}) = C_3$, then $L_b = \mathbb{Q}(\theta_1)$ is a totally-real cyclic cubic field of degree 3.*

3. *If* $\Delta_3 > 0$ *and* $\mathrm{Gal}(b/\mathbb{Q}) = S_3$*, then* $L_b = \mathbb{Q}(\theta_1, \sqrt{\Delta_3})$ *is a totally-real non-cyclic sextic field.*

The singular points of the associated rank-three Heun-type ODE for K *all lie in* L_b*.*

Remark 6. Unlike the PCF-1 case where the CM field is always $\mathbb{Q}(\sqrt{\Delta_2})$, in the cubic case the splitting field is sextic generically and only quadratic-imaginary as a subfield in case (1). The "CM" label therefore extends honestly only to case (1) ($\Delta_3 < 0$, S_3 complex); cases (2) and (3) are totally real and the analogous arithmetic obstruction, if it exists, must be formulated via the cubic Hilbert class field of L_b rather than via imaginary quadratic CM theory. This is the most substantial structural departure from PCF-1.

Conjecture 7 (B_3 part (iii): cubic Stokes signal)**.** *For a degree-three PCF in the irreducible-cubic-CM stratum (case (1) of Conjecture 5:* $\Delta_3 < 0$*), there exists a one-parameter deformation* $K(t)$ *that preserves* $\Delta_3 < 0$ *in a neighbourhood of* $t = 0$ *and respects the splitting-field structure, under which the Stokes exponent*

$$S(t) \;=\; \frac{\log|L(t) - L(0)|}{\log|t|}$$

satisfies $\lim_{t\to 0} S(t) \in (0, 1)$*. The conjectural fractional exponent is governed by the local exponent differences of the associated Heun-type cubic ODE.*

Conjecture 8 (B_3 part (iv): Painlevé-hierarchy reduction)**.** *Each degree-three PCF family in the cell of Conjecture 7 (irreducible* $\mathbb{Z}$*-primitive,* $\Delta_3 < 0$*,* $\mathrm{Gal}(b/\mathbb{Q}) = S_3$ *complex) admits a Painlevé reduction in the P-II–P-VI hierarchy. Heuristic alignment with cubic Galois data (cf. Iwasaki–Kimura–Shimomura–Yoshida [10] and Mazzocco [9]) suggests P-IV as the most likely instance, but no commitment to a specific Painlevé equation is made at the program-statement level. The channel question is itself open: PCF-1 Session E has shown that the natural recurrence-parameter deformation channel and the Borel-resummation channel both* fail *for the degree-2* $\Delta < 0$ *families with respect to V_quad's known P-III*(D_6) *reduction; the cubic case is therefore expected to require a new channel construction, not a verbatim lift of either PCF-1 channel.*

Remark 9. The heuristic preference for P-IV (over P-V or P-VI) within the hierarchy rests on three structural observations, each of which is necessary but not sufficient: (i) P-IV's isomonodromic problem has a single irregular singularity of Poincaré rank $\frac{3}{2}$, the minimal rank compatible with cubic-degree growth of b_n; (ii) the Bäcklund-transformation group of P-IV is $A_2^{(1)}$, the affine analogue of the cubic Galois group $A_3 = C_3$; (iii) P-IV admits Hermite/parabolic-cylinder special-function solutions at half-integer parameters, the natural reservoir of explicit content for the totally-real cyclic case (B) of Conjecture 3. None of (i)–(iii) forces P-IV uniquely; the cubic case may equally well fall into a P-V degeneration or a P-VI specialization at an arithmetic point.

4 Sharp WKB exponent identity, cubic case (Conjecture B4)

The four-part Conjecture B_3 above is a structural prediction about *closed-form admissibility.* We now isolate, and provide 50/50 empirical support for, an analytic-side prediction that is *independent of the trichotomy bin*: a sharp form of the WKB exponent identity for $\log|\delta_n|$ at $d = 3$.

Conjecture 10 (B4: sharp WKB exponent identity)**.** *For every degree-d PCF* $(1, b)$ *of generic type in scope (in the sense of Definition 1 and the regularity hypotheses of PCF-1 v1.3 §5), the convergent residual* $\delta_n = p_n/q_n - L$ *has leading WKB asymptotic*

$$\log|\delta_n| \;=\; -A\,n\log n \;+\; \alpha\,n \;-\; \beta\,\log n \;+\; \gamma \;+\; o(1) \qquad (n \to \infty), \tag{5}$$

with the sharp identification

$$\boxed{A \;=\; 2d.}$$

In particular, $A = 6$ for every cubic PCF in scope.

Relation to PCF-1 v1.3 Theorem 5 ($d = 2$ base case). PCF-1 v1.3 Theorem 5 (*WKB exponent identity, degree two*) proves the form (5) at $d = 2$ with $A \in \{3, 4\}$ split by $\mathrm{sgn}(\Delta_2)$: $A = 3$ for the five $\Delta_2 < 0$ QL families and $A = 4$ for V_quad ($\Delta_2 = -11$). Read alongside Conjecture B4, that $d = 2$ result is simultaneously the *base case* (the identity holds with rational A) and the locus of an anomaly that does not persist at $d = 3$: at $d = 2$ the exponent *splits* between $\{2d - 1, 2d\} = \{3, 4\}$ along $\mathrm{sgn}(\Delta)$, while at $d = 3$ it is *unsplit* at $A = 2d = 6$ across all 50 cubic families harvested below. The $d = 2$ split is therefore a feature specific to degree two; see Open Problem 2.

Empirical harvest, full $d = 3$ coverage. The full 50-row WKB-exponent harvest, combining Sessions A2 (conductor-7 anchor, family 46), B (the 37 $-_S_3_$CM families) and C1 (the 13 non-CM cubic families $+_S_3_$real and $+_C_3_$real), is reproduced as Table 1 below. WKB exponent harvest, Sessions A2 (conductor-7 anchor) + B (37 CM families) + C1 (13 non-CM families).

Summary statistics. Across the 50 cubic families: $A_{\mathrm{fit}} \in [5.85, 6.02]$, with mean $\bar{A} = 5.978$ and standard deviation $\sigma_A = 0.026$. Every family is within 0.15 of $A = 6$; *no* family lies within 0.10 of $A = 5$. The bin-by-bin verdict counts are:

4.1 A failed refinement and what it tells us

Before Session C1, an alternative *refined* form of B4 was on the table:

Conjecture 11 (B4′, FALSIFIED at $d = 3$)**.** *For a degree-d PCF $(1, b)$ of generic type in scope, the WKB exponent of equation (5) satisfies*

$$A_{\mathrm{fit}} = \begin{cases} 2d - 1, & \textit{if the trichotomy bin is elementary-positive} \\ & \quad (+_S_d_\mathrm{real} \;\textit{or}\; +_C_d_\mathrm{real}), \\ 2d, & \textit{otherwise.} \end{cases}$$

The motivation was the $d = 2$ split of PCF-1 v1.3 Theorem 5 ($A \in \{3, 4\}$ along $\mathrm{sgn}(\Delta_2)$) extrapolated naively to the cubic trichotomy bin.

Falsification (Session C1, 2026-05-01). Session C1 ran the WKB fit on the 13 elementary-positive cubic families (the union of $+_S_3_$real and $+_C_3_$real) at dps = 800, fit window $n \in [10, 100]$, step 3. The result is unambiguous: **0** of the 13 families approach the predicted branch $A = 2d - 1 = 5$. All 13 sit in $A_{\mathrm{fit}} \in [5.884, 5.989]$, with mean 5.973 and standard deviation 0.026, indistinguishable from the $-_S_3_$CM bin at $\bar{A} = 5.982$, $\sigma = 0.025$ (Session B). The trichotomy bin is therefore **not** the invariant controlling A at $d = 3$: B4′ is empirically falsified.

What survives. The unsplit form, Conjecture 10 above, survives. The negative result motivates Open Problems 2 and 3 below: *either* the $d = 2$ split is a low-degree artefact and B4 (unsplit, $A = 2d$) holds for all $d \geq 3$, *or* a finer cubic invariant – not the trichotomy bin – splits A at $d = 3$ in a way the present catalogue's coefficient window cannot detect. Both possibilities are testable; see §5.

AEAL note. Documenting failed refinements is part of the AEAL evidence-class methodology of PCF-1 v1.3 §7 [1]: a refined conjecture tested and falsified at the level of a 13-family harvest is a near-miss-class evidence artefact, recorded in the bridge as `sessions/2026-05-01/PCF2-SESSION-C1/` and not promoted beyond its present status as a refuted hypothesis. We do *not* claim B4′ as a "publishable counterexample"; we claim only that B4′ has been ruled out as a candidate refinement of B4.

5 Open Problems

We collect the open problems most directly motivated by the empirical findings of Sessions A2, B, and C1, and by the channel question raised in Conjecture 8.

Open Problem 1 (B4 at $d \geq 4$)**.** *Test whether Conjecture 10 (unsplit, $A = 2d$) extends to $d \geq 4$. A quartic harvest analogous to Sessions B+C1 is the immediate next step: enumerate $\mathbb{Z}$-primitive irreducible quartics in a bounded-coefficient lex window, classify their Galois group (S_4, A_4, D_4, V_4, C_4) and sign of the quartic discriminant, and fit A from the WKB form (5). The prediction is $A = 8$ unsplit; any deviation from 8 falsifies B4 at $d = 4$ and identifies the controlling invariant.*

Open Problem 2 ($d = 2$ anomaly)**.** *Explain why PCF-1 v1.3 Theorem 5 has a* $\mathrm{sgn}(\Delta_2)$*-driven split $A \in \{3, 4\}$ at $d = 2$, while the degree-3 case (Section 4) is* unsplit *at $A = 6$. Two natural hypotheses: (a) the split is a low-degree artefact – e.g. the indicial-root resonance of the rank-two recurrence at $d = 2$ – and Conjecture 10 (unsplit) holds for all $d \geq 3$; (b) the $d = 2$ split is governed by an invariant that becomes vacuous or constant at $d = 3$ (a candidate is the parity of the analytic-rank component of Δ in the indicial expansion). Problem 1 is the cleanest experimental probe between (a) and (b).*

Open Problem 3 (Finer cubic split)**.** *If any cubic invariant – Galois group, field signature, sign of α_3, conductor parity, class number, Stark-unit data, or other –* does *split A at $d = 3$, identify it. Sessions B and C1 have ruled out the trichotomy bin (CM versus $+_S_3_$real versus $+_C_3_$real) as a splitting invariant: the entire 50-family combined harvest sits at $\bar{A} \approx 5.978$, $\sigma \approx 0.026$, with no detected structure correlating $A_{\mathrm{fit}} - 6$ to any of the listed arithmetic invariants. Currently no finer splitting invariant is known.*

Open Problem 4 (Channel construction for the cubic Painlevé reduction)**.** *Construct an isomonodromic-deformation channel under which the cubic $\Delta_3 < 0$ stratum of Conjecture 8 reduces to a Painlevé equation in the P-II–P-VI hierarchy. PCF-1 Session E and the BOREL-CHANNEL-K-SCAN sweep [1] have shown that the recurrence-parameter deformation channel and the Borel resummation channel both fail at $d = 2$ for the non-V_quad $\Delta < 0$ families, so a verbatim lift of either PCF-1 channel is not expected to succeed at $d = 3$. The operationalisation is tracked under the SIARC "CHANNEL-CC-PIPELINE" research track (Prompt F).*

6 Worked Landmarks: Apéry, Catalan-Cubic, Cyclic-C_3

To make the trichotomy of Conjecture 3 concrete we record three landmark families that anchor the three Galois-group cells. None is an empirical result of PCF-2; each is a known PCF whose role under the proposed predicate is fixed in advance, so that Sessions A–C can be calibrated against them.

Calibration role. Table 3 is a calibration anchor, not a results table: each row records what the predicate *must* say if Conjecture 3 is to be consistent with literature. The Catalan-cubic and cyclic-C_3 rows are the positive-test equivalents for B3(i) cells (A) and (B); the Apéry

row, despite its historical centrality at $d = 3$, is *out of scope* of B3(i) by reducibility and thus contributes a separate calibration role discussed in Remark 12.

Remark 12 (Apéry is reducible, hence outside B3(i)). The Apéry denominator $b(n) = 34n^3 + 51n^2+27n+5$ factors as $(2n+1)(17n^2+17n+5)$, so it does not satisfy the irreducibility hypothesis of Conjecture 3. Apéry's recurrence sits in the *out-of-scope, splittable-cubic* bin of trichotomy B3(i), and its known identification $\lim K = 6/\zeta(3)$ [14, 7, 8] is consistent with the restriction of Conjecture 3 to irreducible $\mathbb{Z}$-primitive b: a reducible b admits an algebraic factorization of the recurrence and is not predicted to obstruct a closed form. The CM-style subfield $\mathbb{Q}(\sqrt{-51})$ from the quadratic factor records the arithmetic flavour but does not contradict the predicate. Operationally, Session A's calibration anchor is therefore the Apéry family run through the existing PCF-1 PSLQ pipeline as a *positive control* (target relation $L \sim \zeta(3)$ should be detected) for infrastructure integrity, while the predicate B3(i) itself is tested on the irreducible-only enumeration.

Cubic Galois-group quick checks. For the irreducible $\mathbb{Z}$-primitive case, $\mathrm{Gal}(b/\mathbb{Q})$ is cyclic (C_3) iff $\mathrm{disc}(b)$ is a perfect square in $\mathbb{Q}$ (equivalently in $\mathbb{Z}$ for $\mathbb{Z}$-primitive b); otherwise it is S_3. Examples:

- $b(n) = n^3 - 3n + 1$: disc $= 81 = 9^2$, C_3.
- $b(n) = n^3 + n + 1$: disc $= -31$, S_3 (complex case).
- $b(n) = n^3-2$: disc $= -108$, S_3 (complex case, but the splitting field $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$ is the canonical S_3 cubic of cyclotomic origin).
- $b(n) = n^3 - n - 1$: disc $= -23$, S_3 (complex case, Lehmer's cubic).

The square-discriminant filter applied to $\mathbb{Z}$-primitive irreducible cubics with $|\alpha_i| \leq 5$ is heuristically rare (cyclic cubic fields are sparse in the natural Davenport–Heilbronn / Bhargava counts; see [15] §6.4 for the relevant background on cubic discriminants and Galois groups), and the $\Delta_3 < 0$ fraction is roughly 50% by sign symmetry of the discriminant. The Session A enumeration is expected to populate cells (A) (S_3 complex) and (C) (S_3 totally real) of Conjecture 3 densely, with cell (B) (cyclic) populated sparsely; the calibration of these counts against a finite-coefficient enumeration is itself part of the Session A deliverable.

7 Computational Plan

The computational plan reuses the PCF-1 infrastructure: the Wallis forward-recurrence routine, the PSLQ wrapper with the 18-element basis of PCF-1, the Stokes-exponent diagnostic from PCF-1 §3, and the AEAL evidence-class scaffolding. The plan is organized as three concrete sessions.

PCF-2 Session A: Cubic family enumeration and Δ_3 classification. Enumerate $\mathbb{Z}$-primitive irreducible cubics $b(n) = \alpha_3 n^3 + \alpha_2 n^2 + \alpha_1 n + \alpha_0$ with bounded coefficients $|\alpha_i| \leq 5$ (target: ~ 200 families after gcd reduction and irreducibility filtering). For each family, compute Δ_3, classify the Galois group via the $\sqrt{\Delta_3} \in \mathbb{Q}$ test, and tabulate the splitting type. Produce a stratification table analogous to PCF-1 Table 1 [1], with columns Δ_3, $\sqrt{\Delta_3} \in \mathbb{Q}$?, splitting type, Galois group, real-root count. Estimated effort: 1 week.

PCF-2 Session B: PSLQ scan on irreducible-cubic-CM families. For each family in the $\Delta_3 < 0$ stratum produced by Session A (expected: ~ 80 families), compute $L = K(a_n, b_n)$ to ≥ 320 stable digits via the Wallis recurrence at depth 2500, and run PSLQ against the

PCF-1 18-element basis at working precision 220 digits and integer bound 10^{10}. Confirm or refute Conjecture 3(A). For the $\sqrt{\Delta_3} \in \mathbb{Q}$ subset of $\Delta_3 > 0$ families (C_3 totally-real cyclic), run an extended PSLQ with the basis augmented by cubic-period candidates (Catalan-style Γ-quotients at $1/3, 2/3$; $\sqrt[3]{2}$; $\sqrt[3]{3}$; ${}_2F_1$ values at $1/3$; $\zeta(3)$ already in the base basis) at integer bound 10^{12} and working precision ≥ 320 digits. The targeted detection rate for C_3 families is bounded above by the literature on cubic-period identifications and may be substantially below 1. Estimated effort: 2 weeks.

PCF-2 Session C: Stokes pipeline for cubic case. For 6–10 families flagged in Session B as PSLQ-non-detected, design a Galois-respecting one-parameter deformation, compute $L(t)$ at 11 t-values with the same precision/depth budget as the PCF-1 QL01 analysis, and compute $S(t)$ and $\sigma(t)$ via the formulas of PCF-1 §3. Confirm or refute Conjecture 7. For the strongest Stokes-signal candidate, attempt a P-IV ansatz fit at the same overdetermined-design level used for V_quad in PCF-1, and record the residual matrix. Estimated effort: 2–3 weeks.

Precision and depth budget. The PCF-1 budget (working precision 220 for PSLQ, 320 for the limit; depth 2500; integer bound 10^{10}) extends without modification to $d = 3$ for the PSLQ stage, since the basis size and the expected coefficient bound do not change. The limit-computation budget should be scrutinized at the outset of Session A: cubic b_n amplifies floating-point error growth in the Wallis recurrence by an extra factor of n, so the depth-2500 stable-digit count may drop from ≥ 320 (PCF-1) to ~ 280 at $d = 3$ unless precision is raised correspondingly. We allocate working precision 360 (limit) and 280 (PSLQ) as the conservative defaults, and treat the PCF-1 defaults as fallback.

Risk register. Three risks are identified in advance and each is matched to a mitigation under the AEAL halt-conditions protocol.

AEAL evidence chain. The session chain follows the PCF-1 pattern. Session A is a *formalized* layer (the Galois-group classification is algorithmic and exact). Session B is a *near_miss* layer (each PSLQ non-detection at sufficient precision is a near-miss-class evidence artefact). Session C is a *numerical_identity* layer (the Stokes exponent and any P-IV parameter fit are numerical-identity-class artefacts). *Independently_verified* promotion requires cross-session reproduction of $L(0)$ to ≥ 50 digits across at least two independent sessions, identical to the PCF-1 cross-session protocol.

8 Why This Matters

PCF-2 is positioned to deliver one of two outcomes, both informative.

Outcome U (universality). If Conjecture 3 is confirmed across the Session B sweep, the Δ-discriminant transcendence predicate of PCF-1 lifts cleanly to higher degree, with the cubic Galois-group axis as the natural refinement. This would be the first non-trivial empirical evidence that the discriminant predicate is degree-general rather than a degree-two coincidence, and it would provide a roadmap for PCF-3 ($d = 4$, with the quartic Galois groups S_4, A_4, D_4, V_4, C_4) and beyond. The universality outcome also strengthens the analytic prediction of Conjecture 8: a Painlevé equation per degree, indexed by the Galois group, would be a striking new entry in the PCF–Painlevé correspondence catalogued by [9].

Outcome S (specificity). If Conjecture 3(A) fails — in particular, if any $\Delta_3 < 0$ family yields a non-trivial PSLQ relation against the PCF-1 basis — then degree two is identified as a special boundary at which the discriminant predicate works, and the higher-degree picture is controlled by a different invariant (Picard–Fuchs operator structure, modular-monodromy condition, or a cubic-period basis that supersedes the PCF-1 18-element basis). The Apéry family ($a_n = n^6$, $b_n = 34n^3 - 51n^2 + 27n - 5$) is the historical exemplar of a $d = 3$ PCF whose limit $\zeta(3)/6$ is detected against a cubic-period basis; PCF-2 would then be repositioned as a search for the natural generalization of the Apéry basis. This outcome is no less valuable: it sharpens the conjectural content of PCF-1 by identifying the precise scope of the discriminant predicate.

Either way, a stable scope-anchor. The role of this program statement is to fix the scope, the conventions, and the success criteria *before* the computational sweep begins. The AEAL methodology requires that conjectures be stated explicitly and that evidence-class promotion follow a documented chain; this document discharges the first half of that requirement. Sessions A–C of §7 discharge the second half.

9 Relation to Existing Literature

The mathematical content of PCF-2 sits at the intersection of three literatures, none of which addresses the cubic PCF transcendence question directly.

Painlevé and number theory. Mazzocco's lecture notes on Painlevé equations and number theory [9] catalogue the algebraic-solution loci of P-VI (and by reduction, P-V, P-IV, P-III) and identify a small set of arithmetic Painlevé parameters at which classical special-function solutions exist. The PCF-1 V_quad result lifts naturally into this framework as a P-III(D_6) solution at a specific $\mathbb{Q}(\sqrt{-11})$-arithmetic point. The cubic analogue (Conjecture 10) predicts that P-IV — whose isomonodromic problem has Poincaré rank 3/2 at the irregular singularity, the natural cubic refinement of P-III's rank 1 — governs the cubic case. Mazzocco's Backlund-transformation analysis of P-IV at $A_2^{(1)}$ parameters is the structural object PCF-2 Session C is designed to intersect.

Iwasaki–Kimura–Shimomura–Yoshida and the Gauss-to-Painlevé arc. The monograph [10] traces the lineage from the Gauss hypergeometric equation to the Painlevé transcendents via isomonodromic deformation. Each rank-two Heun reduction of a Painlevé equation has a dedicated treatment, and the cubic-Galois-group classification of Schwarz-triangle / Takeuchi groups appears explicitly. The PCF-1 framework borrows the Spec(K) τ_{arith} component from this lineage; PCF-2 extends it to the cubic Galois-group axis.

Apéry's irrationality proof and the cubic PCF exemplar. Van der Poorten's account [14] of Apéry's proof [7] of the irrationality of $\zeta(3)$ is the historical landmark of cubic PCFs. The Apéry recurrence is a $d = 3$ PCF whose limit is $6/\zeta(3)$, identified against a basis that contains $\zeta(3)$ explicitly; this is the unique $d = 3$ family in the literature that is fully resolved against a closed-form basis known in advance. Beukers [8] reformulated the proof via a Picard–Fuchs operator on a one-parameter family of K3 surfaces. The Picard–Fuchs framing is the candidate alternative predicate raised under outcome S of §8: if the discriminant predicate fails at $d = 3$, the natural competitor is "L is identifiable iff the associated Picard–Fuchs operator is a pullback of a hypergeometric operator over $\mathbb{Q}$". PCF-2 Session B is the test that decides between these two predicates.

Ramanujan-Machine search and PSLQ infrastructure. The Ramanujan-Machine programme [12] produced the bulk of the PCF candidate database used by PCF-1, and its PSLQ-based identification pipeline is what PCF-1 inherits. The Ferguson–Bailey PSLQ algorithm [13] is basis-dependent: a closed-form identification is only as strong as the basis it is checked against. The cubic-period augmentation of the PCF-1 basis (§7, Session B) is therefore not optional but mandatory for the C_3 cell of Conjecture 3; the $\Gamma(1/3)$, $\Gamma(2/3)$, $\sqrt[3]{2,3}$, ${}_2F_1(\cdot;1/3)$ augmentation is the smallest basis under which any reasonable C_3-cyclic limit could conceivably be detected.

Cohen, cubic fields, and the heuristic distribution. Cohen's textbook treatment of cubic fields and their discriminants [15] provides the standard background for the heuristic distribution of $\mathrm{Gal}(b/\mathbb{Q})$ across $\mathbb{Z}$-primitive irreducible cubics: cyclic cubics are sparse (Davenport–Heilbronn / Bhargava counts), which is consistent with the $\sim$ 5–10% figure used in §6 as a Session A pre-estimate. PCF-2 does not rely on a published PCF-specific cubic enumeration; the Session A enumeration of §7 performs its own bounded-coefficient sweep and reports the empirical distribution as a deliverable.

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Table 1: Combined 50-row WKB-exponent cubic harvest (Sessions B + C1). A_{fit} fitted from $\log|\delta_n| \sim -A\,n\log n + \alpha n - \beta\log n + \gamma$ at dps 800. 'pred.' is the Conjecture B4′ branch: $2d-1=5$ for the elementary-positive bins $+_S_3$_real and $+_C_3$_real, $2d=6$ for $-_S_3$_CM. Out-of-scope rows are reducible controls or non-cubic.

ID	$(\alpha_3,\alpha_2,\alpha_1,\alpha_0)$	**bin**	Δ_3	**pred.**	A_{fit}	**stderr**	**verdict**
1	(1,-3,-3,-3)	$-_S_3$_CM	-864	6	5.9678	0.0019	✓
2	(1,-3,-3,-2)	$-_S_3$_CM	-459	6	5.9726	0.0027	✓
3	(1,-3,-3,-1)	$-_S_3$_CM	-108	6	5.9697	0.0018	✓
4	(1,-3,-3,2)	$+_S_3$_real	621	5	5.9719	0.0018	ambiguous
5	(1,-3,-3,3)	$+_S_3$_real	756	5	5.9715	0.0016	ambiguous
6	(1,-3,-2,-3)	$-_S_3$_CM	-823	6	5.9536	0.0098	✓
7	(1,-3,-2,-2)	$-_S_3$_CM	-472	6	5.9688	0.0029	✓
8	(1,-3,-2,-1)	$-_S_3$_CM	-175	6	5.9695	0.0028	✓
9	(1,-3,-2,1)	$+_S_3$_real	257	5	5.9774	0.0017	ambiguous
10	(1,-3,-2,3)	$+_S_3$_real	473	5	5.9770	0.0014	ambiguous
11	(1,-3,-1,-3)	$-_S_3$_CM	-716	6	5.9783	0.0014	✓
12	(1,-3,-1,-2)	$-_S_3$_CM	-419	6	5.9690	0.0050	✓
13	(1,-3,-1,-1)	$-_S_3$_CM	-176	6	5.9778	0.0014	✓
14	(1,-3,-1,1)	$+_S_3$_real	148	5	5.9813	0.0012	ambiguous
15	(1,-3,-1,2)	$+_S_3$_real	229	5	5.9818	0.0012	ambiguous
16	(1,-3,0,-3)	$-_S_3$_CM	-567	6	5.9788	0.0024	✓
17	(1,-3,0,-2)	$-_S_3$_CM	-324	6	5.9884	0.0025	✓
18	(1,-3,0,-1)	$-_S_3$_CM	-135	6	5.9837	0.0010	✓
19	(1,-3,0,1)	$+_C_3$_real	81	5	5.9860	0.0009	ambiguous
20	(1,-3,0,3)	$+_C_3$_real	81	5	5.9870	0.0008	ambiguous
21	(1,-3,1,-2)	$-_S_3$_CM	-211	6	5.9891	0.0007	✓
22	(1,-3,1,-1)	$-_S_3$_CM	-76	6	5.9904	0.0009	✓
23	(1,-3,1,3)	$-_S_3$_CM	-76	6	5.9827	0.0043	✓
24	(1,-3,2,-3)	$-_S_3$_CM	-239	6	6.0151	0.0109	✓
25	(1,-3,2,-2)	$-_S_3$_CM	-104	6	5.9884	0.0026	✓
26	(1,-3,2,-1)	$-_S_3$_CM	-23	6	5.9960	0.0010	✓
27	(1,-3,2,1)	$-_S_3$_CM	-23	6	5.9823	0.0064	✓
28	(1,-3,2,2)	$-_S_3$_CM	-104	6	5.9957	0.0002	✓
29	(1,-3,2,3)	$-_S_3$_CM	-239	6	5.9962	0.0002	✓
30	(1,-3,3,-3)	$-_S_3$_CM	-108	6	6.0241	0.0130	✓
31	(1,-3,3,1)	$-_S_3$_CM	-108	6	5.8516	0.0736	ambiguous
32	(1,-3,3,2)	$-_S_3$_CM	-243	6	6.0005	0.0001	✓
33	(1,-3,3,3)	$-_S_3$_CM	-432	6	6.0010	0.0001	✓
34	(1,-2,-3,-3)	$-_S_3$_CM	-519	6	5.9781	0.0014	✓
35	(1,-2,-3,-2)	$-_S_3$_CM	-244	6	5.9786	0.0013	✓
36	(1,-2,-3,-1)	$-_S_3$_CM	-23	6	5.9791	0.0013	✓
37	(1,-2,-3,1)	$+_S_3$_real	257	5	5.8836	0.0479	ambiguous
38	(1,-2,-3,2)	$+_S_3$_real	316	5	5.9806	0.0012	ambiguous
39	(1,-2,-3,3)	$+_S_3$_real	321	5	5.9810	0.0011	ambiguous
40	(1,-2,-2,-2)	$-_S_3$_CM	-268	6	5.9831	0.0011	✓
41	(1,-2,-2,-1)	$-_S_3$_CM	-83	6	5.9835	0.0010	✓
42	(1,-2,-2,2)	$+_S_3$_real	148	5	5.9850	0.0009	ambiguous
43	(1,-2,-1,-3)	$-_S_3$_CM	-439	6	5.9870	0.0008	✓
44	(1,-2,-1,-2)	$-_S_3$_CM	-236	6	5.9875	0.0008	✓
45	(1,-2,-1,-1)	$-_S_3$_CM	-87	6	5.9879	0.0007	✓
46	(1,-2,-1,1)	$+_C_3$_real	49	5	5.9889	0.0007	ambiguous
47	(1,-2,-1,3)	$-_S_3$_CM	-31	6	5.9899	0.0006	✓
48	(1,-2,0,-3)	$-_S_3$_CM	-339	6	5.9913	0.0006	✓
49	(1,-2,0,-2)	$-_S_3$_CM	-172	6	5.9918	0.0005	✓
50	(1,-2,0,-1)	$-_S_3$_CM	-59	6	5.9923	0.0005	✓

Bin	Families	In-band ($\lvert A_{\text{fit}} - 6\rvert \le 0.15$)	Closer to $A = 2d - 1 = 5$?
$-$_S_3_CM (B)	37	37/37	0/37
$+$_S_3_real (C1)	10	10/10	0/10
$+$_C_3_real (C1)	3	3/3	0/3
Combined cubic catalogue	**50**	**50/50**	**0/50**

Table 2: Per-bin verdicts of Conjecture 10 at $d = 3$. The 10 $+$_S_3_real entries comprise the families with $\Delta_3 > 0$ and $\sqrt{\Delta_3} \notin \mathbb{Q}$ in the lex window of §7 Session A; the 3 $+$_C_3_real entries are the totally-real cyclic cubics, of which family 46 ($b = n^3 - 2n^2 - n + 1$, $\Delta_3 = 49 = 7^2$) is the conductor-7 anchor of Session A2. Reducible-control families (e.g. Apéry's $b = 34n^3 + 51n^2 + 27n + 5$) are out-of-scope of B$_3$(i) by Remark 12 and are not included in Table 1.

Landmark	$b(n)$	Δ_3	Reducible?	Gal($b/\mathbb{Q}$)	Predicted cell
Apéry $\zeta(3)$	$34n^3 + 51n^2 + 27n + 5$	$-459 = -3^3 \cdot 17$	yes: $(2n + 1)(17n^2 + 17n + 5)$	C_2 on $\mathbb{Q}(\sqrt{-51})$	out-of-scope (see Rmk. 12)
Catalan-cubic seed	$n^3 + n + 1$	-31	no	S_3 (complex)	A: PSLQ-null + Stokes
Cyclic-C_3 probe	$n^3 - 3n + 1$	$81 = 9^2$	no	C_3	B: closed-form candidate

Table 3: Three landmark $b(n)$. The Apéry row uses the recurrence denominator $b(n) = 34n^3 + 51n^2 + 27n + 5$ from [14, 7, 8]; an exact symbolic computation (sympy, exact integer arithmetic) gives $\Delta_3 = \mathrm{disc}(b) = -459 = -3^3 \cdot 17$, and the polynomial *factors* over $\mathbb{Q}$ as $(2n + 1)(17n^2 + 17n + 5)$. Hence b is *reducible*, the splitting field is $\mathbb{Q}(\sqrt{17^2 - 4 \cdot 17 \cdot 5}) = \mathbb{Q}(\sqrt{-51})$, and the Galois group of b over $\mathbb{Q}$ is C_2 rather than C_3 or S_3. The Catalan-cubic seed $b(n) = n^3 + n + 1$ is the simplest irreducible cubic with negative discriminant (disc $= -31$, Gal $= S_3$, splitting field $\mathbb{Q}(\theta, \sqrt{-31})$); it is the cleanest test of Conjecture 3(A). The cyclic-C_3 probe $b(n) = n^3 - 3n + 1$ is the standard simplest totally-real cyclic cubic with disc $= 81 = 9^2$; it is the cleanest test of Conjecture 3(B). The accompanying a_n for each landmark is fixed in Session A by the smallest $\mathbb{Z}$-primitive choice consistent with the balanced-class convention (2).

Risk	Likelihood	Mitigation
Wallis recurrence loses precision faster than expected at $d = 3$, so PSLQ-null results are not actually null but precision-starved.	Medium.	Run Apéry landmark first; if $\zeta(3)/6$ is not detected at the conservative budget, escalate dps to 500 and depth to 4000.
Cubic-period basis is too small: C_3-cyclic limits identify against constants outside the augmented basis.	Medium–high.	Treat all C_3 "no relation" as preliminary; flag for a follow-up cubic-Hilbert-class-field basis sweep before claim promotion.
P-IV ansatz fits all fail in Session C, mirroring the QL01 outcome of PCF-1.	High (this is what happened in the $d = 2$ case).	Fall back to the Stokes-exponent $S < 1$ predicate alone, weakening Conjecture 8 to a fractional-power existence claim, exactly as PCF-1 v1.2 did for QL01–QL26.

Table 4: Risk register for the PCF-2 computational plan.